AIN SHAMS UNIVERSITY — MSc LITERATURE DEFENSE & ROADMAP

Biomechanical & Neuromuscular Hand Grasp Synergies

From High-DoF Biological Redundancy to Minimal-Actuator Prosthetic & Exoskeleton Control

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Supervision Committee: Prof. Dr. Mohamed Hamdy Abdelhafez & Dr. Mohamed Ibrahim Awad (Ain Shams University)
Literature Synthesized: 18 Peer-Reviewed Papers & 2026 Benchmarks (2019–2026) | KIN-MUS UJI, Ninapro DB2 & Encord EMG-1
Core Mathematical Anchors: PCA (Kinematic), NMF (Muscular), MMF (Cross-Domain), JGKSE (ORQ/IRQ > 90%) & Simscape Multibody

Problem Statement & Ain Shams Institutional Baseline

Bridging the Curse of Dimensionality in Hand Biomechanics & Clinical Rehabilitation

21–27 DoFs
Human Hand Kinematics
Driven by 34+ muscles

33 Grasps
Feix Grasp Taxonomy
Full ADL precision & power repertoire

85% ± 5.1%
Ain Shams Glove (2021)
Sarwat, Maged & Awad (Sensors 2021)

K_min ≤ 4
Target Actuation Bound
VAF ≥ 90% + Force-Closure Stability

1. The Engineering & Clinical Bottleneck

- Fully actuated dexterous hands (16–22 motors) suffer from high weight, cost, cognitive load, and clinical abandonment.
- Naive 1-DoF grippers fail multi-axis precision grasps and induce compensatory shoulder/trunk kinematics.
- Purely global PCA reduction drops critical individual joints (thumb opposition & palm arch CMC).
- Raw sEMG amplitude decoders degrade under electrode drift and grasp force scaling (Low/Med/High).

2. Upgrading the Ain Shams Data Glove (Sarwat & Awad, 2021)

- **Department Baseline (MDPI Sensors 2021, Dr. Mohamed I. Awad & Dr. Shady Maged):** 10 flex sensors (MCP/PIP), 5 fingertip FSRs, 1 6-axis IMU.
- Achieved **85% (±5.1%)** post-stroke home rehabilitation assessment accuracy via cloud ML classifiers.
- **Thesis Evolution:** Upgrade from raw 16-channel passive classification into a unified neuro-mechanical synergy architecture.
- Replace ad-hoc sensor mapping with mathematically invariant PCA, NMF, MMF, and JGKSE manifolds validated in Simscape.

Eng. Mohamed Zedan — Prof. Dr. Mohamed Hamdy & Dr. Mohamed I. Awad

Supervisors' Core Directives & Research Questions

Strict Mathematical & Biomechanical Framing (Prof. Dr. Mohamed Hamdy & Dr. Mohamed I. Awad)

RQ1: Cross-Domain Kinematic–Muscular Coupling

- **Research Question:** What is the biomechanical & mathematical relationship linking joint Kinematic Synergies (S_kin) and forearm Muscle Synergies (W_EMG) across Feix grasp taxonomies?
- `PCA: Q - Q_bar = U * Sigma * V^T ≈ S_K * C_K`
- `NMF: min || E_EMG - W * H ||_F^2, W ≥ 0, H ≥ 0`
- **MMF Bridge (Scano et al., 2023):** Jointly factorizes signed kinematic angles and non-negative sEMG envelopes into shared neural activations **H_shared**.

RQ2: Minimal Actuators & Mechanical Stability

- **Research Question:** What is the minimum number of actuators / DoFs (**K_min ≤ 4**) reproducing daily grasps with **VAF ≥ 90%** and high mechanical grasp stability?
- `VAF(K) = (1 - ||Q - Q_hat||_F^2 / ||Q||_F^2) * 100% ≥ 90%`
- **JGKSE Local Fidelity (Chu et al., 2026):** Enforce Overall **ORQ > 90%** AND Individual **IRQ_i > 90%** for ≥ 90% of joints (protecting thumb & CMC).
- **Stability Proof:** Force-closure & slip testing in **Simulink / Simscape Multibody**.

Strict Scope Constraints & Multi-Dataset Benchmarks

- Zero "gadget / graduation project" framing — purely focused on neuro-biomechanical synergy extraction, dimensionality bounds, and dynamic force-closure validation.
- **Datasets: KIN-MUS UJI** (28 subjects, 26 ADLs, 18 CyberGlove DoFs + 7 sEMG — locally staged), **Ninapro DB2** (28 subjects, 20 grasps via MMF), and **Encord EMG-1** (100 hrs total; 8-ch 2000 Hz sEMG + 120 Hz OptiTrack).

Pillar 1: Kinematic Synergies & Subspace Invariance

Proving Low-Dimensional Linear Subspaces Across Large Cohorts & Dexterous Tasks

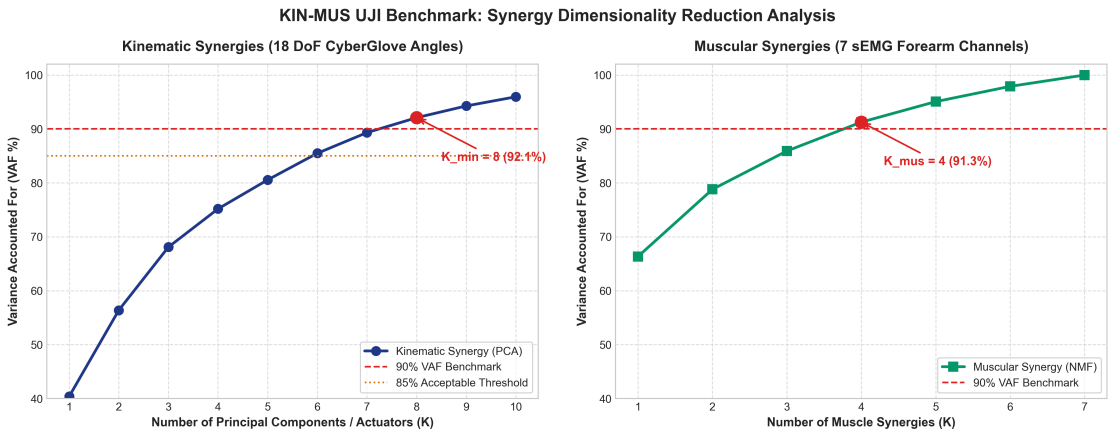
Study & Venue	Cohort & Acquisition Setup	Method	Verified Dimensionality & Variance Findings
Jarque-Bou et al. (2019) *JNER*	77 subjects, 20 grasps, 17-DoF CyberGlove II	PCA kinematic synergies	First **3 PCs > 50%** variance; **12 PCs > 80%** variance across 77 subjects. Defines universal pad/palmar & thumb opposition primitives.
Bazina et al. (2024) *MDPI Robotics*	**KIN-MUS UJI**: 23 functional ADLs, 16 finger flexion DoFs	Inter-joint linear coupling (30 clusters)	Reduces 16 flexion/extension DoFs by up to **11 DoFs**, leaving **5–15 independent DoFs (median 12)** across ADLs.
West & Hogan (2024) *IEEE BioRob*	7 subjects (MIT IRB), 19-DoF CyberGlove @ 200 Hz	PCA + VAF grasp vs. wire-harness manipulation	First **2 PCs > 80%** variance in reach-to-grasp; active tool manipulation recruits higher DoFs atop the shared grasp core.
West & Hogan (2025) *IEEE ICRA*	20-DoF: 7 wire-harness + 5 expert pianists (20 yrs exp.)	**VAF-CI** & Subspace Principal Angles	Core **3–4 kinematic synergies** span a statistically invariant subspace at **VAF ≥ 90%** (<5% marginal gain) — **Anchors K_min ≤ 4**.

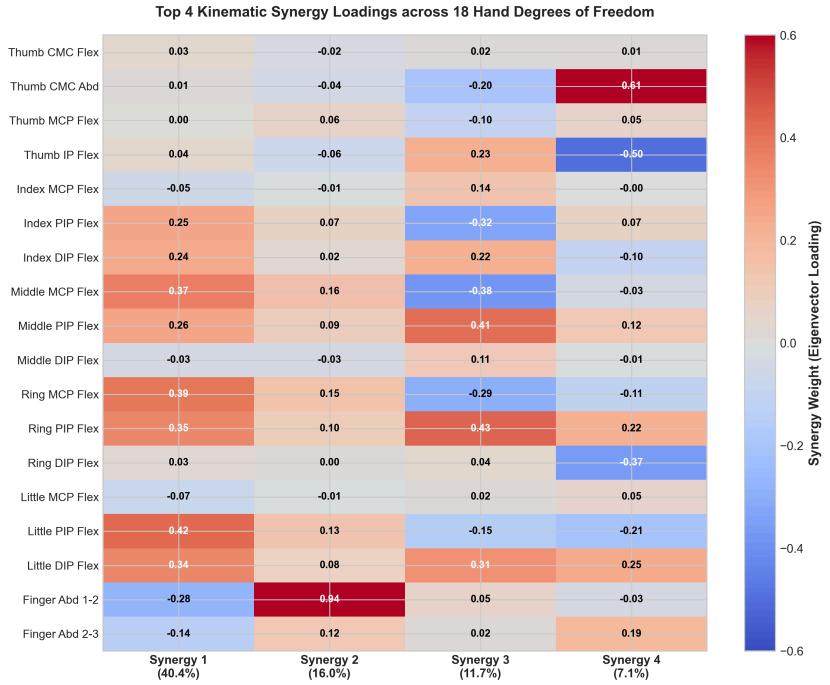
Core Takeaway for RQ2

- Across both industrial tool use and expert classical piano playing, West & Hogan (2025) prove mathematically via VAF Confidence Intervals that **3 to 4 synergies** form the conserved low-dimensional basis of human hand control.

Empirical Baseline Analysis on Local KIN-MUS UJI Dataset

Direct PCA Extraction Across 28 Subjects, 26 Daily Activities (ADLs) & 18 Joint DoFs





Cumulative VAF Scree Saturation

- Steep variance capture within **K = 3 to 4 synergies** across all 18 CyberGlove joints; diminishing marginal return beyond K = 4.

Anatomical Synergy Coupling Weights

- **PC1–PC2:** Coordinated MCP/PIP flexion across digits 2–5 with thumb flexion; **PC3–PC4:** Thumb opposition & abduction modulation.

Pillar 2: Muscle Synergies (NMF) & Cross-Domain MMF Coupling

Linking Forearm sEMG Motor Modules to Kinematic Posture Under Variable Forces (RQ1 Core)

Teng et al. (2021) — IEEE TNSRE

- 6 hand grasps tested across 3 force levels (Low, Medium, High).
- **Intra-force accuracy:** 90.2%–91.4% across all 6 schemes.
- **Overall accuracy: 85.8%** (MSC + DFTR).
- **Cross-force generalization:** NMF MSC (**78.1%–78.7%**) significantly beat LDA (**68.4%–76.9%**, $p < 0.05$).
- Proves NMF synergies remain stable when grip force changes.

Scano et al. (2023) — IEEE Access ★ RQ1

- **Ninapro DB2:** 28 healthy subjects, 20 grasps (6 repetitions).
- **Mixed Matrix Factorization (MMF):** Jointly factorizes signed joint angles and non-negative sEMG envelopes.
- Extracted **7 unified kinematic-muscular synergies** per subject achieving mean reconstruction **$R^2 > 0.75$**.
- Direct mathematical engine answering **RQ1**.

Meattini (2023) & Olikkal (2022)

- **Meattini et al. (2023, IEEE RA-L):** Self-supervised NMF (**$W \in R^{8\times2}$**) + DNN mapping 8-ch sEMG to 3 grasp primitives (AR10 hand) without MoCap labels.
- **Metrics: R^2 = 0.8423, nMSE = 0.1355, RMSE = 0.0735**.
- **Olikkal (2022) & Yeung/Farina (2022):** SVD/Tensor synergies + Kalman Filtering suppress sEMG drift.

Pillar 3: Underactuated Hardware & Joint-Group Synergies (JGKSE)

Comparing K=1 SoftHand, K=2 Mini X-Hand & 2026 SX-Hand (33/33 Feix Taxonomy Coverage)

Metric / Architecture	Li et al. (2022) BPI SoftHand	Sun et al. (2022) Mini X-Hand	Chu et al. (2026) SX-Hand [T-RO]
Actuators (K)	**K = 1 actuator**	**K = 2 actuators**	**K = 12 actuators** (synergy-grouped)
Joints & Palm	15 joints / 19 DoFs (~£65 3D-printed)	5 fingers + differential palm mechanism	21 joints incl. movable Ring/Little **CMC palm**
Synergy Formulation	Adaptive + Soft Synergies (series springs)	Mechanical embedding of 2 Eigengrasps	**JGKSE:** ORQ > 90% & IRQ > 90% (≥90% joints)
Verified Performance	Passive shape adaptation via single drive	Reproduces **29 / 33 Feix grasps (~88%)**	**ORQ=96.3±1.5%**, min **IRQ=94.6±2.7%** **33/33 Feix (100%)**, **10/10 Kapandji**, **20 kg**

Critical Engineering Bridge Between Sun (2022) and Chu (2026)

- **Why K=2 stops at 29/33 Feix grasps:** Global PCA sacrifices thumb opposition (responsible for >40% of hand function) and palm arch CMC compliance.
- **Our RQ2 Strategy:** Combine primary flexion eigengrasps with JGKSE-isolated thumb/CMC actuation to achieve full Feix dexterity within **K_min ≤ 4**.

Musculoskeletal Modeling & Simscape Dynamic Force Validation

Why Kinematic Synergies Alone Fail Under Load — Integrating Lapresa et al. (2022 & 2024)

Lapresa et al. (2022) — OpenSim Baseline

- Validated the OpenSim **ARMS Hand_Wrist_Model (26 DoFs)** during human reach-to-grasp trials.
- Achieved optical motion-capture Inverse Kinematics marker tracking error **< 5 mm**.
- Provides anatomically verified joint axes, moment arms, and tendon excursion constraints before mechanical design.

Lapresa et al. (2024) — Force Synergies & Simscape

- Combined Vicon kinematics & finger contact pressure sensors across 8 subjects and 21 reach-to-grasp tasks.
- LDA classifier separated power vs. precision grasps with **89% accuracy**.
- **Simulink & Simscape Multibody:** Proved dynamic grasp stability increases significantly when combining kinematic & force synergies.

Thesis Dynamic Verification Protocol (Meeting Prof. Dr. Mohamed Hamdy's Stability Criterion)

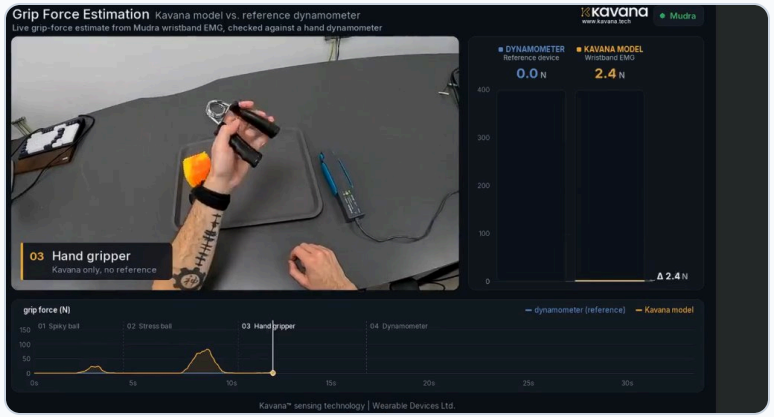
- Evaluate our **K_min ≤ 4** JGKSE + MMF controller inside **Simulink / Simscape Multibody** under realistic object masses, friction coefficients, and disturbance wrenches.
- Confirm zero slip / object drop across canonical Feix power and precision test primitives at **VAF ≥ 90%**.

2026 Breakthrough Benchmark: Encord EMG-1 & Kavana Wristband

100-Hour Open sEMG–MoCap Dataset Verified Locally on Take OP-001/s003/02







100 Hours
EMG-1 Benchmark
8ch-mocap: 12 subj, 920 takes @ 2kHz

MAE 4.21°
vs. 9.02° CV Baseline
53.3% lower error across 20 DoFs

100% Immune
Visual Occlusion
Continuous pose when vision blind

0–80 N
Isometric Grip Force
Wrist EMG vs hand dynamometer

Verified Local Dataset Integration

- Cloned `encord-team/EMG-1` & verified take `OP-001/s003/02` (`take.h5` 16.93 MB, `take.c3d` 1.34 MB, `automatic_est.h5` 2.88 MB): 8-ch 2000 Hz sEMG + 23 OptiTrack markers @ 120 Hz.

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Temporal Synergy Primitives & 2026 VLA Foundation Models

From PCA + ProMPs (Pham et al., 2026) to Latent Action Bottlenecks (NVIDIA GR00T 2, Nov 2026)

Pham et al. (2026) — Robot. Auton. Syst. (PCA + ProMPs)

- Recorded 20-DoF haptic exoskeleton glove across 13 cleaning tasks (5 train, 8 unseen), deployed on Shadow Hand.
- Combined spatial **PCA** with temporal **Probabilistic Movement Primitives (ProMPs)** inside the synergy subspace.
- **Verified Metrics: 5 PCs (15 DoFs)** & **7 PCs (20 DoFs)** capture **≥ 90% variance** with **< 10% reconstruction error**.
- Nearest-expert ProMP selection generalized cleanly to 8 unseen tasks.

NVIDIA GEAR Lab (Nov 2026) — GR00T 2 Whitepaper

- 135-page architecture whitepaper scaling Vision-Language-Action (VLA) models across humanoid & 20–22 DoF dexterous hands.
- Combines **VQ-VAE Latent Action Quantization** with **Conditional Flow Matching** action transformers.
- **Architectural Lesson:** Even large-scale foundation policies depend on a **low-dimensional latent synergy bottleneck** to avoid compounding errors during contact-rich manipulation.

Convergence Across Hardware & AI

- Classical underactuated mechanisms ($K_{min} \leq 4$) and 2026 robotics foundation models converge on the exact same mathematical truth: control happens in a compressed synergy manifold.

18-Paper Master Taxonomy Synthesis & Identified Research Gaps

Pinpointing Exactly Where Existing Literature Stops and Where Our MSc Thesis Begins

Pillar	Representative Works	State-of-the-Art Result	Open Literature Gap We Solve
1. Kinematic & Sim	Jarque-Bou 19, Lapresa 22/24, Bazina 24, West & Hogan 24/25	Invariant 3–4D subspace (VAF ≥ 90%); OpenSim <5mm; 16→5–15 DoFs	Disconnected from sEMG muscle coupling & minimal physical actuator mapping
2. Muscular & MMF	Teng 21, Yeung 22, Olikkal 22, Scano 23, Meattini 23, EMG-1 26	Force-robust NMF (85.8%); MMF 7 synergies (R^2>0.75); MAE 4.21°	MMF synergies not compressed into **K_min ≤ 4** tendon topology
3. Hardware	Li 22 (K=1), Sun 22 (K=2), Chu 26 (K=12 SX-Hand)	K=2 gives 88% Feix; JGKSE+CMC gives 100% Feix at K=12	Gap between K=2 (88%) & K=12 (100%) leaves **K_min ≤ 4** unexplored
4. ASU Base & AI	Sarwat & Awad 21, Pham 26, NVIDIA GR00T 2	ASU Glove (85% rehab); ProMPs & VQ-VAE synergy bottlenecks	ASU Glove uses raw 16-sensor ML; needs MMF + JGKSE + Simscape upgrade

Unified Ain Shams MSc Thesis Novelty Statement

- First unified framework integrating **MMF cross-domain coupling (RQ1)** + **JGKSE (ORQ > 90%, IRQ > 90% with Palm CMC)** + **Simscape Multibody Force-Closure** inside **K_min ≤ 4 actuators (RQ2)**.

Proposed MSc Thesis Execution Roadmap (4 Phases)

Actionable Engineering Pipeline from Mathematical Extraction to Ain Shams Prototype Validation

Phase 1: Cross-Domain Math

- Run PCA, NMF & MMF on local **KIN-MUS UJI** (28 subj, 26 ADLs) & **Encord EMG-1** (8ch-mocap).
- Map kinematic-muscular coupling weights across Feix power vs. precision grasps (RQ1).
- Compute VAF-CI & Principal Angles.

Phase 2: K_min ≤ 4 Optimization

- Apply **JGKSE** (Chu 2026) enforcing both **ORQ > 90%** & **IRQ_i > 90%**.
- Derive optimal K ≤ 4 allocation: Power Flexion + Precision Pinch + Thumb/CMC Arch.
- Verify full Feix reachability.

Phase 3: Simscape Dynamics

- Construct parametric 21-DoF hand in **Simulink / Simscape Multibody**.
- Simulate K_min ≤ 4 tendon routing & series springs under load (Lapresa 2024).
- Prove dynamic force closure & anti-slip stability.

Phase 4: ASU Integration

- Integrate synergy decoder onto **Ain Shams Data Glove** pipeline (Sarwat & Awad 2021).
- Benchmark real-time latency & force tracking error.
- Compile & defend MSc Thesis.

Ready for Committee Approval & Phase 1 Execution

- All 18 baseline works are fact-checked, synthesized, and aligned with **KIN-MUS UJI** & **Encord EMG-1** locally — ready for immediate Phase 1 MMF & JGKSE experimentation.